



Increased Risk of Dementia in Patients with Craniofacial Trauma: A Nationwide Population-Based Cohort Study

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OBJECTIVE: The role of maxillofacial trauma in dementia risk is not well established. The aim of this study was to evaluate the association between craniofacial trauma, including facial bone fracture and traumatic brain injury (TBI), and dementia.

METHODS: Using Taiwan's National Health Insurance Research Database, we identified 501,889 adults who had had ≥ 1 medical record of craniofacial trauma between 2000 and 2010 and did not have a dementia diagnosis at baseline. Diagnoses of craniofacial trauma, including facial bone fracture and TBI, and dementia were made using *International Classification of Diseases, Ninth Revision* codes. The standardized incidence ratio was used to determine whether craniofacial trauma was associated with a greater risk of incident dementia compared with the general population. The Cox proportional hazards model was used to predict the risk of dementia among the trauma cohort by comparing the patients with and without comorbidities.

RESULTS: A total of 501,889 patients with craniofacial trauma were included, of which 1.5% ($n = 7804$) developed dementia. Facial bone fracture (standardized incidence ratio, 1.58; 95% confidence interval, 1.25–2.00) was shown to be associated with an increased dementia risk compared with the general population. In addition,

craniofacial trauma accompanied with postinjury comorbidities was associated with an increased risk of dementia during follow-up periods compared with the group without comorbidities.

CONCLUSIONS: Craniofacial traumas, especially facial bone fracture, were associated with an increased risk of subsequent dementia. Maintaining a high index of suspicion for associated TBIs in all patients with facial trauma is crucial, even if no obvious initial signs and symptoms of brain injury are observed.

INTRODUCTION

Many traumatic brain injuries (TBIs) occur worldwide each year. The estimated annual global incidence of TBIs requiring attention or resulting in hospitalization or death is >10 million.¹ In the United States, approximately 1.4 million people have TBI every year.² In Taiwan, the estimated annual number of TBIs is 52,000, up to 25% of which are fatal.³ Mild TBI is estimated to comprise 90% of all cases of TBI.⁴ TBI leads to high mortality and incurs enormous economic costs.^{5–7} TBI constitutes a major global health and socioeconomic problem, with neurobehavioral sequelae contributing to long-term disability.⁸ One of the most critical long-term disabilities of

Key words

- Facial bone fracture
- Mild traumatic brain injury
- Neurodegenerative disease
- Postinjury comorbidities
- Posttraumatic epilepsy
- Traumatic brain injury

Abbreviations and Acronyms

AD: Alzheimer disease

CI: Confidence interval

CNS: Central nervous system

CTE: Chronic traumatic encephalopathy

EAR: Excess absolute risk

ED: Emergency department

HR: Hazard ratio

ICD-9-CM: International Classification of Diseases, Ninth Revision, Clinical Modification

NHIRD: National Health Insurance Research Database

PTE: Posttraumatic epilepsy

SIR: Standardized incidence ratio

TBI: Traumatic brain injury

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TBIs is dementia because multiple epidemiologic studies have shown that experiencing a TBI in early or midlife is associated with an increased risk of dementia in later life.⁹

A recent meta-analysis reported that head injury is associated with increased risks of dementia.¹⁰ Accumulating epidemiologic evidence shows that moderate to severe TBI is a crucial risk factor for dementia.^{10–12} A systematic review showed that Alzheimer disease (AD) was associated with moderate to severe TBI but not with mild TBI unless loss of consciousness occurred.¹² Relatively few studies have assessed the association specifically between mild TBI and dementia.^{13–15} These studies have suggested that there is likely a small, but substantial, risk of dementia after a single mild TBI, which is not solely the result of reverse causation or confounding factors¹⁵; however, the association is not well established.

Maxillofacial traumas affect many patients with trauma and may lead to life-threatening conditions, such as airway compromise and profuse blood loss. Because of the anatomic proximity of the facial skeleton and cranium, patients with maxillofacial trauma may be at an increased risk for head injuries.

We hypothesized that facial bone fractures combine with undetectable or delayed brain injury, namely neurologic injury, and are associated with an increased risk of subsequent dementia. According to our review of the relevant literature, no previous large population-based studies have investigated the association between facial bone fracture and the risk of subsequent dementia. In our study, we retrospectively analyzed data from the Taiwan National Health Insurance Research Database (NHIRD) to clarify any association between craniofacial trauma, including TBI and facial bone fracture, and risk of subsequent dementia during a follow-up period.

In addition, to analyze confounding factors during the follow-up period, we investigated the role of postinjury comorbidities, including cerebrovascular disease, posttraumatic epilepsy (PTE), and central nervous system (CNS) infection, in the association between craniofacial trauma and dementia. We hypothesized that patients with craniofacial trauma with postinjury comorbidities are at a higher risk of dementia compared with those without these comorbidities.

METHODS

Study Population

A craniofacial trauma cohort was established by including all patients who had received a diagnosis of facial bone fracture or TBI from the inpatient and outpatient emergency department (ED) visit medical claims filed between 2000 and 2010 in Taiwan. The data source in this study was the NHIRD, which routinely collected health information from all Taiwan National Health Insurance (NHI) beneficiaries from 1995 until 2017. The NHI is a single-payer system that requires participation of all Taiwan residents and covered 99.6% of the population in the NHIRD at the end of 2014.¹⁶ The NHIRD includes all encounter records (dates of inpatient and outpatient visits, medical diagnoses, medical expenditures, prescription details, operations, and procedures) as well as demographic information (sex, date of birth, and place of residence) of each patient. Every individual has a unique encrypted personal identification number, which makes

it possible to link different files. The International Classification of Diseases, Ninth Revision, Clinical Modification (ICD-9-CM) was used in NHIRD as a diagnostic coding system. This study was approved by the institutional review board of the Chang Gung Memorial Hospital, and because the data used were encrypted, the study was exempt from obtaining informed consent from the patients.

Ascertainment of Facial Bone Fracture and TBI

TBI was defined using the U.S. Centers for Disease Control and Prevention criteria¹⁷ and identified from medical records of patients with ICD-9-CM codes of 800.0–801.9, 803.0–804.9, 850.0–854.1, or 959.01. The facial bone fracture cohort was identified from patients who had received the diagnosis code 802 during hospitalization or ED visit. We divided patients with TBI into 2 groups (mild TBI and moderate to severe TBI), on the basis of ICD-9-CM codes. Patients with diagnosis codes of 800.0, 800.5, 801.0, 801.5, 803.0, 803.5, 804.0, 804.5, 850.0, 850.1, 850.5, and 850.9 (with a fifth digit of 0, 1, 2, 6, 9, or missing) or 854.0 (with a fifth digit of 1, 2, 6, 9, or missing) were defined as the mild TBI group and the others were defined as the moderate to severe TBI group.

To obtain clearer understanding of the effects of a recent TBI sustained in middle-aged or older adulthood on the risk of development of dementia, patients aged 55 years or older without missing sex and date of birth information were included in the cohort if they were diagnosed as having a TBI or facial bone fracture during an inpatient or ED visit during 2000–2010. We excluded patients who had a diagnosis of dementia before craniofacial trauma occurred.

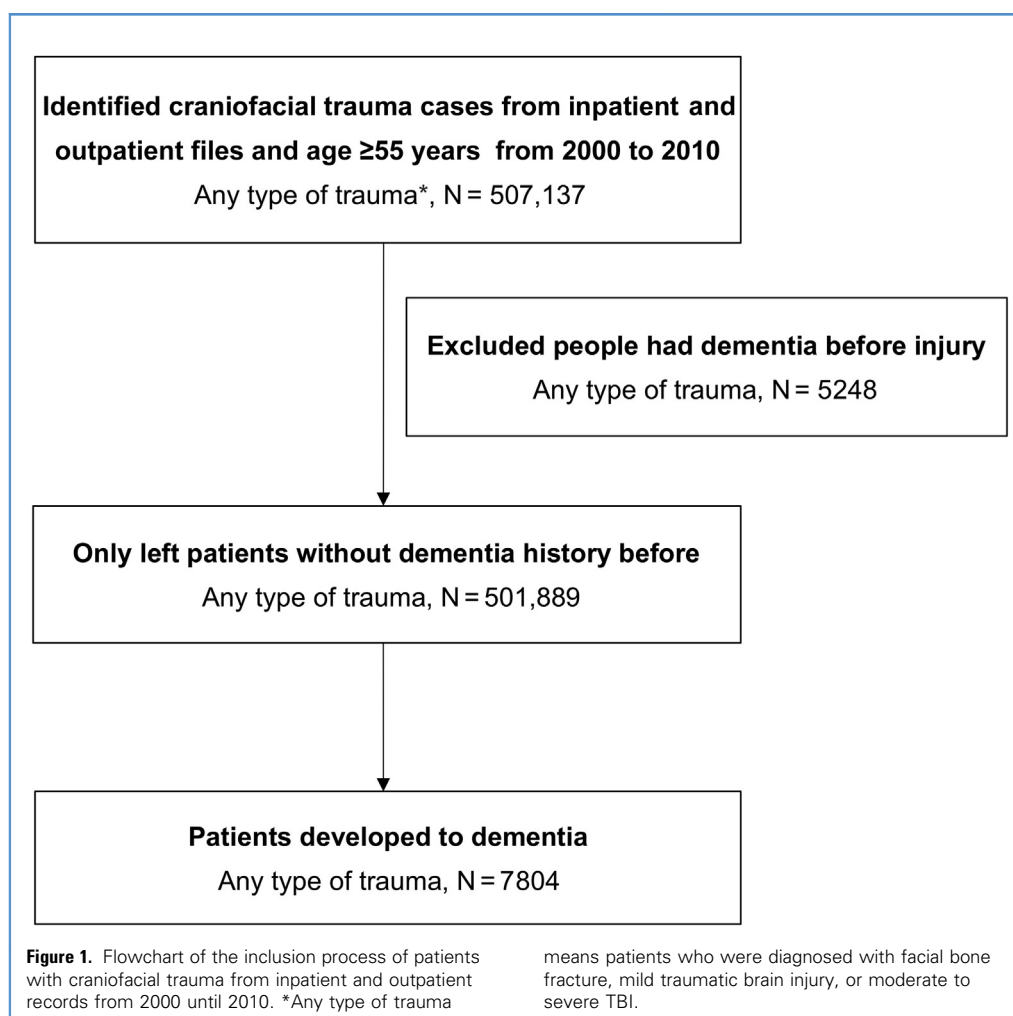
We divided patients into 2 groups on the basis of a combination of different injury types. The first group included patients with only a single type of craniofacial trauma during the study period, which were facial bone fracture, mild TBI, and moderate to severe TBI. The second group included patients with combined different types of injury during the same period, which were facial bone fracture and mild TBI; facial bone fracture and moderate to severe TBI; mild TBI and moderate to severe TBI; and facial bone fracture, mild TBI, and moderate to severe TBI.

Ascertainment of Dementia

The outcome of interest was dementia, which was identified from the Taiwan Catastrophic Illness Registry (ICD-9-CM codes 290 and 294) from 2000 to 2010. The Catastrophic Illness Certificate is a system designed to help ease the medical burden of patients with severe disease. Disease-associated diagnostic information is sent to the insurance administration for review by commissioned expert panels before approval of waivers. This registry contains information on these patients with unique personal identification, diagnosis, and application date.

Statistical Analysis

The standardized incidence ratio (SIR) was used to estimate the risk of dementia in the craniofacial trauma cohort compared with the general population. We used observed incidences of dementia divided by the expected incidences of dementia to obtain the SIRs based on age-specific incidence rates in 5-year age intervals of the general population. The expected incidences of dementia were computed using



age-specific rates from a reference population, weighted according to the age structure of the study population. The incidence rates of dementia in the general population were calculated using the NHIRD. The incident rates were obtained using the new cases of dementia divided by the person-years of the specific population in each age group in the same calendar year. The at-risk cohort comprised patients without dementia diagnosis, and the sum of those patients' time from the date of injury to dementia occurrence, death, deregistration, or December 31, 2010, was calculated to obtain the person-years. We also subtracted the incidence rate of the craniofacial trauma cohort from the incidence rate of the general population to obtain the excess absolute risk (EAR) of dementia risk. The 95% confidence intervals (CIs) for the SIR and EAR were calculated based on the assumption of Poisson distribution.¹⁸ Kaplan-Meier plots were used to estimate the cumulative incidence of dementia. To compare the cumulative incidence between the craniofacial trauma cohort and the general population, we matched craniofacial trauma cases with controls at a ratio of 1:1 by age and sex. The first date of craniofacial trauma was defined as the index date, and we

assigned this index date to each matched control. The Cox proportional hazards model was used to predict the risk of dementia among the injury cohort by comparing the patients with and without comorbidities (CNS infection, PTE, and cerebrovascular disease). Only patients whose comorbidities occurred after the diagnosis of craniofacial trauma were included in this definition. All tests of statistical hypothesis were performed on the 2-sided 5% level of significance. All analyses were performed using SAS version 9.4 (SAS Institute, Cary, North Carolina, USA).

RESULTS

After excluding patients with a dementia diagnosis before the initial craniofacial trauma, the cohort included 501,889 patients with craniofacial trauma. Patients were followed up from the date they were diagnosed with craniofacial trauma, including TBI or facial bone fracture, until they were diagnosed with dementia. A total of 7804 patients were diagnosed with dementia during the follow-up period (Figure 1). Baseline patient characteristics were reported in Table 1.

Table 1. Characteristics of Patients with Different Types of Craniofacial Trauma from 2000 to 2010

Variables	Facial Bone Fracture	Mild Traumatic Brain Injury	Moderate to Severe Traumatic Brain Injury
Number	23,890	399,058	149,475
Age, n (%)			
55–64 years	9990 (41.82)	136,821 (34.29)	44,353 (29.67)
65–74 years	7482 (31.32)	126,017 (31.58)	49,627 (33.2)
75–84 years	4986 (20.87)	103,134 (25.84)	42,620 (28.51)
85+ years	1432 (5.99)	33,086 (8.29)	12,875 (8.61)
Age (years), mean $\pm$ standard deviation	68.7 $\pm$ 9.50	70.5 $\pm$ 9.86	71.40 $\pm$ 9.61
Sex, n (%)			
Male	15,083 (63.14)	207,527 (52.00)	88,585 (59.26)
Female	8,807 (36.86)	191,531 (48.00)	60,890 (40.74)
Place of residence, n (%)			
Urban	5393 (22.57)	88,392 (22.15)	34,247 (22.91)
Suburban	6229 (26.07)	102,906 (25.79)	38,253 (25.59)
Rural	11,748 (49.18)	198,756 (49.81)	73,716 (49.32)
Unknown	520 (2.18)	9004 (2.26)	3259 (2.18)
Income levels, n (%)			
Quintile 1	4586 (19.20)	83,897 (21.02)	33,421 (22.36)
Quintile 2	2525 (10.57)	41,394 (10.37)	15,747 (10.53)
Quintile 3	5491 (22.98)	102,762 (25.75)	39,198 (26.22)
Quintile 4	6382 (26.71)	89,531 (22.44)	30,376 (20.32)
Quintile 5	4804 (20.11)	79,530 (19.93)	29,932 (20.02)
Unknown	102 (0.43)	1944 (0.49)	801 (0.54)
Occupation, n (%)			
Dependents of the insured individuals	6983 (29.23)	123,308 (30.90)	45,539 (30.47)
Civil servants, teachers, military personnel, and veterans	1686 (7.06)	29,420 (7.37)	10,890 (7.29)
Nonmanual workers and professionals	2027 (8.48)	28,023 (7.02)	10,102 (6.76)
Manual workers	10,316 (43.18)	170,015 (42.60)	62,892 (42.08)
Other	2878 (12.05)	48,292 (12.10)	20,052 (13.41)

During follow-up, 23,890 patients were diagnosed with facial bone fracture at a mean age of 68.7 years. A total of 399,058 patients were diagnosed as having mild TBI and 149,475 patients were diagnosed as having moderate to severe TBI. The mean ages of these patients with TBI were 70.5 and 71.4 years, respectively. Men were predominant among patients with facial bone fracture or TBI. Patients with facial bone fracture or mild TBI were more likely to be in the 55-year to 64-year age group; patients with moderate to severe TBI were more likely to be in the 65-year to 74-year age group (Table 1). Patients described in Table 1 may have multiple diagnoses at the injury time or in their lifetime. For example, patients who were diagnosed with facial bone fracture may have another diagnosis such as mild TBI.

Patients with a single type of craniofacial trauma diagnosis during the follow-up period were associated with an overall doubled risk of dementia compared with the general population (SIR, 2.02; 95% CI, 1.98–2.07). Overall dementia incidence in patients with single-type craniofacial trauma was 37.89 per 10,000 person-years, which corresponded to an EAR of 19.14 (95% CI, 18.54–19.76) per 100,000 person-years. For dementia, the increased risk was the highest for moderate to severe TBI (SIR, 2.18; 95% CI, 2.07–2.30) and lowest for facial bone fracture (SIR, 1.58; 95% CI, 1.25–2.00). However, the risk of dementia in patients with facial bone fracture still remained higher than that in the general population (Table 2).

Table 2. Risk of Dementia in Single Type of Craniofacial Trauma

	All*	Facial Bone Fracture†	Mild Traumatic Brain Injury‡	Moderate to Severe Traumatic Brain Injury§
Observed number	7589	71	4451	1409
Expected number	3754.92	44.85	2496.95	645.83
Standardized incidence ratio (95% confidence interval)	2.02 (1.98–2.07)	1.58 (1.25–2.00)	1.78 (1.73–1.84)	2.18 (2.07–2.30)
Incidence/10,000				
Observed	37.89	23.48	32.83	43.89
Expected	18.75	14.83	18.42	20.12
Excess absolute risk/10,000 person-years (95% confidence interval)	19.14 (18.54–19.76)	8.60 (5.62–12.6)	14.41 (13.78–15.06)	23.77 (22.11–25.51)

*During 2,002,673.81 person-years.
†During 30,241.05 person-years.
‡During 1,355,913.88 person-years.
§During 321,044.88 person-years.

Further sensitivity analyses to assess the robustness of our primary result produced similar findings. **Figure 2** showed the cumulative probability of dementia in patients with craniofacial trauma compared with matched controls in the general population. Patients with facial bone fracture were more likely to be diagnosed as having dementia compared with the matched controls (2.31% vs. 1.06%; $P < 0.01$). Those with TBIs were also more likely to be diagnosed as having dementia compared with the matched controls (mild TBI, 2.75% vs. 1.28%; $P < 0.01$; moderate to severe TBI, 3.19% vs. 1.38%; $P < 0.01$).

Figure 3 showed a significant interaction between craniofacial trauma severity and dementia. The P for log-rank test for 2 groups showed that increasing levels of trauma severity were associated with increasing risk of having dementia (facial bone fracture vs. mild TBI, $P < 0.0029$; mild TBI vs. moderate to severe TBI, $P < 0.0001$; facial bone fracture vs. moderate to severe TBI, $P < 0.0001$). In addition, it showed that the moderate to severe TBI group were most likely and faster to be diagnosed as having dementia among the groups with craniofacial traumas.

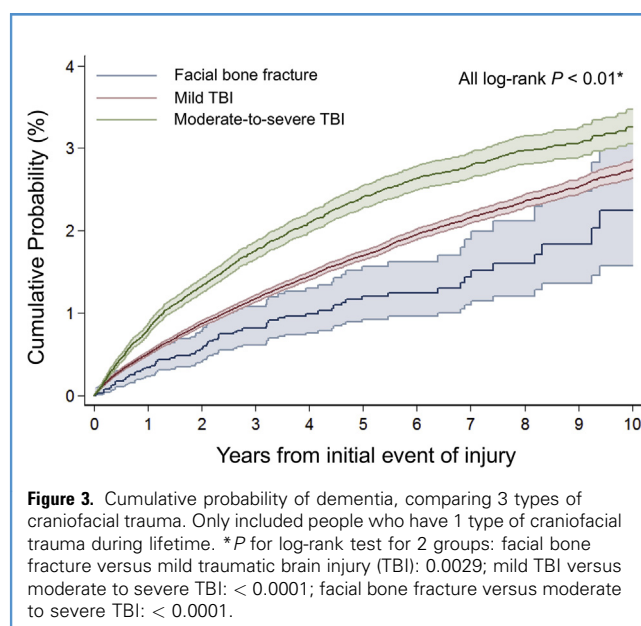
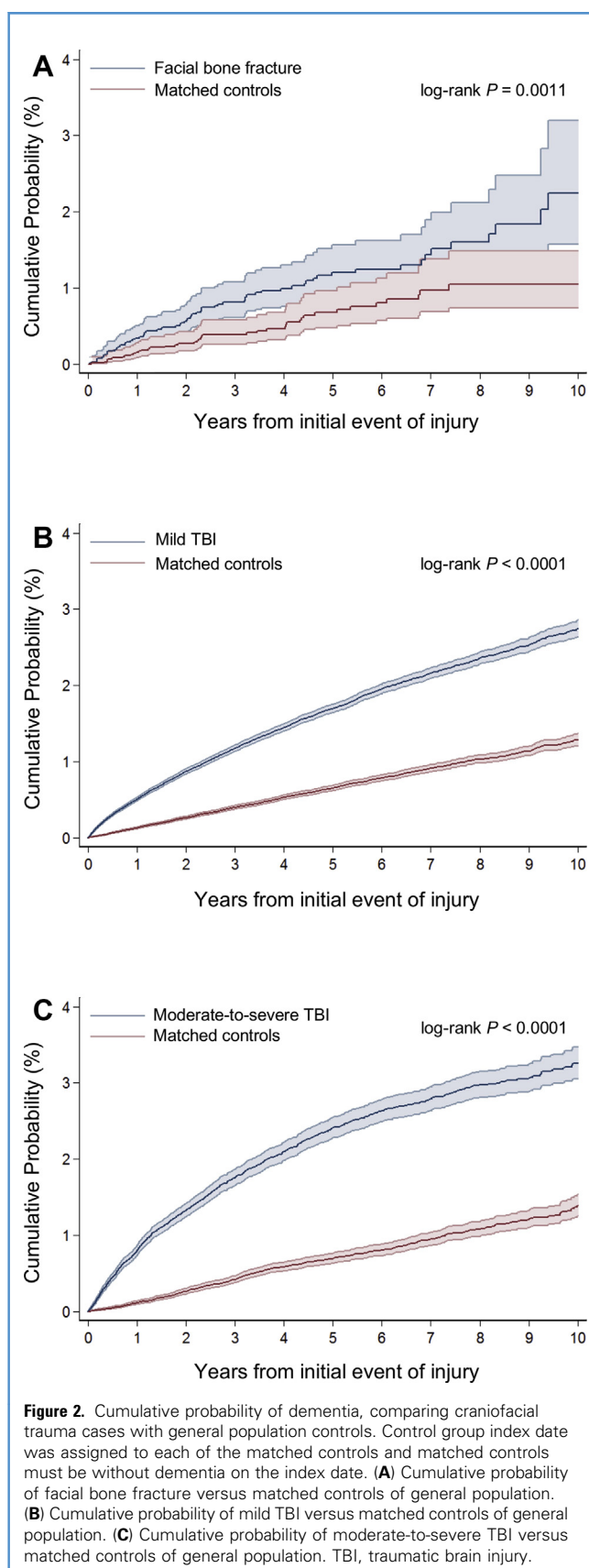
Patients with multiple types of craniofacial trauma during the follow-up period were associated with an increased risk of dementia compared with the general population. Among patients with facial bone fracture and any type of TBI during follow-up, SIRs were significantly increased ($P < 0.01$) (facial bone fracture and mild TBI: SIR, 1.84; 95% CI, 1.52–2.21; facial bone fracture and moderate to severe TBI: SIR, 2.62; 95% CI, 1.97–3.49; facial bone fracture, mild TBI, and moderate to severe TBI: SIR, 3.61; 95% CI, 2.95–4.4). Among patients who had multiple types of TBI (i.e., both mild and moderate to severe TBI), SIRs were also significantly increased ($P < 0.01$) (mild TBI and moderate to severe TBI: SIR, 3.04; 95% CI, 2.88–3.20). For dementia, the increased risk was the highest in patients with facial bone fracture, mild TBI, and moderate to severe TBI (SIR, 3.61; 95% CI, 2.95–4.41) and lowest in patients with facial bone fracture and mild TBI (SIR, 1.84; 95% CI, 1.52–2.21). However, the risk of dementia in patients with facial bone fracture and mild TBI remained higher

than that in the general population (**Table 3**). After excluding all patients with facial bone fracture, we stratified patients based on trauma frequency during the follow-up period and compared these 4 groups of patients with the general population. The risk of dementia after a single TBI, 2 TBIs, 3 TBIs, and ≥ 4 TBIs was increased (single TBI: SIR, 2.02; 95% CI, 1.97–2.07; 2 TBIs: SIR, 1.98; 95% CI, 1.88–2.10; 3 TBIs: SIR, 2.22; 95% CI, 2.00–2.48; ≥ 4 TBIs: SIR, 2.28; 95% CI, 1.94–2.69). However, no significant difference was observed between single TBI and >1 TBI (**Figure 4A**).

In addition, we stratified patients with TBI and facial bone fracture based on trauma frequency during the follow-up period and compared these 4 groups of patients with the general population. The risk of dementia after a single injury event, 2 injury events, 3 injury events, and ≥ 4 injury events was increased (single injury event: SIR, 1.86; 95% CI, 1.81–1.91; 2 injury events: SIR, 1.77; 95% CI, 1.67–1.87; 3 injury events: SIR, 2.01; 95% CI, 1.81–2.24; ≥ 4 injury events: SIR, 2.09; 95% CI, 1.77–2.48). No significant difference between single craniofacial trauma and repetitive craniofacial trauma was observed (>1 injury events) (**Figure 4B**).

We conducted analyses to test confounding factors during the follow-up period, such as postinjury comorbidities (i.e., PTE, cerebrovascular disease, and CNS infection). After patients with these 3 comorbidity diagnoses before craniofacial trauma were excluded, the cohort comprised patients with craniofacial trauma with or without PTE, cerebrovascular disease, or CNS infection during follow-up. In the unadjusted model, patients with craniofacial trauma with these 3 post-injury comorbidities during follow-up had a significantly higher risk of dementia compared with those without these comorbidities (**Table 4**).

Female sex and an age of 65 years or older were strongly associated with an increased risk of dementia in patients with craniofacial trauma. In addition, for dementia, the increased risk was the highest in patients who were 75–84 years old (adjusted hazard ratio [HR], 4.77; 95% CI, 4.38–5.20). Individual



adjustment for each covariate changed the HR by <10% except for PTE and cerebrovascular disease (model adjusted for PTE: HR, 1.37; 95% CI, 1.17–1.59; cerebrovascular disease: adjusted HR, 1.10; 95% CI, 1.02–1.18) (Table 4).

DISCUSSION

Our retrospective study showed that craniofacial trauma, including facial bone fracture and TBI, was associated with an increased risk of dementia. The novel finding of our study was that facial bone fracture was associated with an increased dementia risk compared with the general population. According to our review of the relevant literature, no previous study has assessed the association between facial bone fracture and the risk of subsequent dementia. Moreover, in our study, we included not only patients with TBI but also those with facial bone fracture, with the intention of assessing the association between craniofacial trauma and the risk of subsequent dementia.

Facial Bone Fracture Versus Dementia

Several studies^{19–21} have indicated that traumatic energy is largely absorbed by the facial skeleton, which acts as a cushion in reducing the risk of TBI, thus protecting the intracranial organs. However, other studies^{22–24} have shown that traumatic energy that is sufficient to cause facial injuries has the potential for concomitant brain injuries. Moreover, some recent studies have suggested that forces acting on the face may transmit them directly to the neurocranium, resulting in a more serious brain injury. That is, the presence of facial bone fractures increased the risk of TBI.^{25–27} Hohlrieder et al.²⁶ reported that patients with central midface fractures and skull base fractures are prone to significantly increased risks of intracranial hemorrhage. Zandi and Seyed Hoseini²⁷ reported that facial bone fracture, especially of bones that are in anatomic proximity to the cranium, was a marker for an increased risk of brain injuries.

Table 3. Risk of Dementia in Multiple Types of Craniofacial Trauma

	Facial Bone Fracture + Mild TBI*	Facial Bone Fracture + Moderate to Severe TBI†	Mild TBI + Moderate to Severe TBI‡	Facial Bone Fracture + Mild TBI + Moderate to Severe TBI§
Observed number	110	47	1406	95
Expected confidence interval	59.88	17.91	463.18	26.32
Standardized incidence ratio (95% confidence interval)	1.84 (1.52–2.21)	2.62 (1.97–3.49)	3.04 (2.88–3.20)	3.61 (2.95–4.41)
Incidence/10,000				
Observed	28.09	38.91	61.71	57.94
Expected	15.29	14.83	20.33	16.05
Excess absolute risk/10,000 person-years (95% confidence interval)	12.77 (9.48–16.83)	24.01 (16.09–34.46)	41.39 (38.8–44.11)	42.09 (32.76–53.23)

TBI, traumatic brain injury.
 *During 39,165.48 person-years.
 †During 12,078.87 person-years.
 ‡During 227,834.23 person-years.
 §During 16,395.42 person-years.

Some studies^{27–29} have shown that TBI could be missed or unrecorded in patients with facial bone fracture. Puljula et al.²⁹ found high rates of nonrecording (46%) of TBI in patients with craniofacial fractures. Powell et al.²⁸ reported that the diagnosis of mild TBI was frequently absent from ED medical records despite patients reporting findings consistent with a mild TBI diagnosis when interviewed by study personnel. Zandi and Seyed Hoseini²⁷ indicated that the chance for a head injury to be missed in patients with facial trauma is not low. Specifically, mild TBI frequently remained unrecorded in patients with facial bone fracture.

This situation may be a result of the complexity of the available diagnostic criteria. Nonneurologic specialists, including plastic surgery, otolaryngology, and oral surgery specialists, manage almost half of the patients with facial bone fracture. These specialists are relatively unfamiliar with definitions of mild TBI, which vary in different guidelines and may be confusing, especially to nonneurologic specialists. Therefore, patients with unrecorded mild TBI were significantly less frequently referred for follow-up visits.

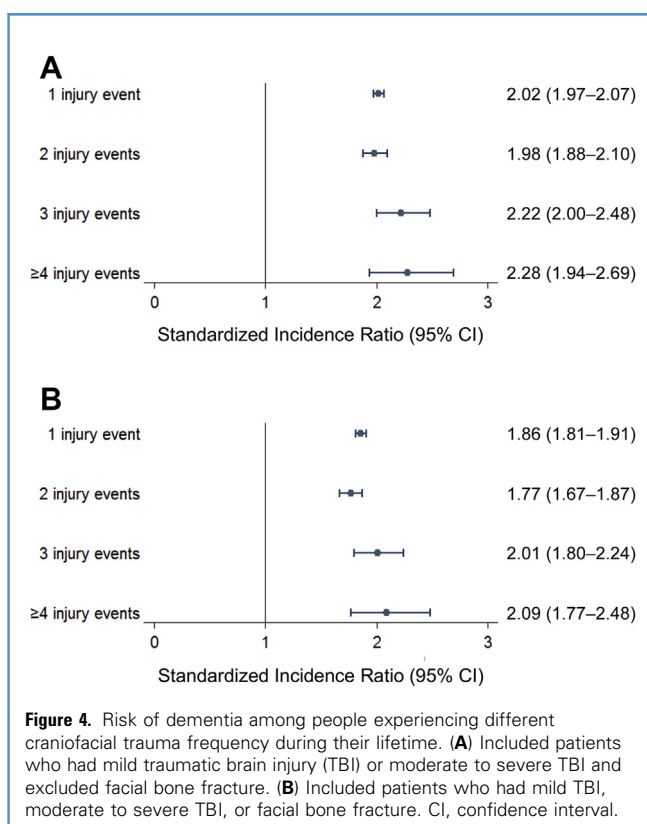
TBI is a highly complex disorder with neurologic injury that involves primary and secondary injury phases.³⁰ Mechanical forces at the moment of TBI cause rapid tissue deformation, resulting in primary physical damage. Secondary injury mechanisms include a wide variety of processes, such as depolarizations and disturbances of ionic homeostasis, release of neurotransmitters, mitochondrial dysfunction, neuronal apoptosis, and initiation of inflammatory and immune responses.³¹ Such secondary injury begins within seconds to minutes after the primary insult and may continue for days, weeks, and months, progressively contributing to deteriorating neurologic function.

We could suppose that the facial skeleton absorbs some, but not all, forces from the brain. When the face is under sufficient magnitude of impact force, the facial struts fail and transmit or transfer the impact to the adjacent neurocranium. In our study, patients who were diagnosed with facial bone fracture may have had an unrecorded diagnosis of brain injury or a delayed brain injury, such as mild TBI, during the follow-up period. This theory may explain why facial bone fracture was associated with an increased risk of dementia in the present study. Thus, to assess the association between brain injury, especially for mild TBI, and subsequent dementia more comprehensively, patients with facial trauma should be included.

The presence of facial trauma should be of clinical concern because it can present as a marker for substantial transfer of energy to the brain, with possible associated brain injury.^{32–35} Maintaining a high index of suspicion for associated TBIs in all patients with facial trauma is crucial, even with no presentation of initial obvious signs and symptoms of brain injury. The increased risk of dementia after facial bone fracture is almost certainly the result of an undiagnosed mild TBI at the time of injury. All patients with craniofacial trauma should be carefully investigated for signs and symptoms of TBI frequently, and the diagnosis of TBI, if present, should be recorded at discharge.

Mild TBI Versus Dementia

Many studies have assessed the risk of dementia after mild TBI. A meta-analysis³⁶ indicated that a history of TBI, including mild TBI, was associated with the development of neurologic and psychiatric illness. Some studies have reported a significant association between mild TBI and dementia.^{37,38} However, these studies were limited by a relatively small number of mild TBI-exposed



patients. Several large studies^{13,39} using national registries of administrative health data have also specifically assessed the risk of dementia after single mild TBI. Lee et al.¹³ reported that mild TBI was associated with a greater than 3-fold increased risk of dementia. However, the average time from mild TBI to dementia diagnosis was only 1 year, which raised the possibility of reverse causation. Gardner et al.³⁹ reported that mild TBI was significantly associated with a 22%–26% increased risk of dementia over a subsequent duration of follow-up (5–7 years), but only among patients aged 65 years and older during TBI.

In our large population-based study, mild TBI was associated with an increased dementia risk compared with the general population (SIR, 1.78; 95% CI, 1.73–1.84), and the median time from mild TBI to dementia was 3.46 years (95% CI, 3.45–3.48 years).

Moderate to Severe TBI Versus Dementia

Accumulating epidemiologic evidence suggests that moderate to severe TBI is a crucial risk factor for neurodegenerative diseases such as AD.^{11,12,40} We found similar results in our large population-based study that moderate to severe TBI was associated with an increased dementia risk compared with the general population. Moreover, moderate to severe TBI caused the highest subsequent dementia risk in patients with craniofacial trauma. This finding might explain the association between TBI severity and dementia. That is, the risk of dementia was greater with increasing TBI severity.

Trauma Severity Versus Dementia

We reported an increasing level of trauma severity from facial bone fracture, mild TBI to moderate to severe TBI was associated with increasing risk of having dementia. Some studies had similar results.^{39,41,42} Gardner et al.³⁹ reported that patients with moderate to severe TBI at 55 years or older or mild TBI at 65 years or older had an increased risk of developing dementia. In addition, more severe TBI is associated with an ever-increasing risk of young-onset dementia.⁴¹ Plassman et al.⁴² showed that there was a dose-response pattern such that risk of dementia progressively increases as the severity of TBI increase.

In addition, our study showed that there was a trend toward a dose-response relationship between craniofacial trauma severity and interval from trauma to diagnosis of dementia. That is, the more severe the trauma severity was, the quicker dementia developed.

Trauma Frequency Versus Dementia

A notable finding in the subgroup analyses in our study was that no significant difference for dementia risk was observed not only between single TBI and repetitive TBI groups but also between single craniofacial trauma and repetitive craniofacial trauma groups. Our review of studies investigating the association between TBI frequency and the risk of neurodegenerative diseases showed that studies have mainly assessed the risk of neurodegenerative diseases, especially chronic traumatic encephalopathy (CTE), after repetitive mild TBI or repetitive subconcussive head injuries, most of which focused on athletes such as boxers and football players.^{43,44} Lehman et al.⁴³ found a 3-fold increased mortality resulting from neurodegenerative causes, particularly for AD and amyotrophic lateral sclerosis, on death certificates among retired National Football League players compared with the general population. However, the study could not show a difference between single TBI and repetitive TBI. Furthermore, the study had a cohort effect; the frequency of mild TBI was higher and injury interval from initial head trauma to subsequent injury was shorter during the lifetime of boxers and football players than in patients with TBI resulting from vehicle accidents or falls. Incomplete recovery time of the previous injury might have played a role in the neurodegeneration after brain injury in the athlete group. Therefore, it was unclear to what degree studies of one cohort of patients consisting of boxers and football players exposed to repetitive mild TBI may be applied to other cohorts.

Gao et al.⁴⁵ reported that the accumulation of brain injuries could lead to more severe neuropathologic and neurobehavioral changes after TBI in animal models. Nevertheless, TBI is heterogeneous, and one model cannot fully recapitulate all the mechanisms that occur in humans. Moreover, rats in the repetitive mild TBI group received 4 head impacts and each hit interval was 24 hours. This interval might be too short and may not be completely applicable to human situations. The effect of accumulation in neurodegeneration after traumatic brain trauma is not well established.

Overall, both single and repetitive TBI are associated with an increased risk of neurodegenerative disease.⁴⁶ According to a previous study,⁴⁷ there may be accumulation effects of TBI on neurodegenerative diseases; however, no well-established evidence is available for the general population. In our study, we

Table 4. Risk of Dementia in Patients with Posttraumatic Epilepsy, Cerebrovascular Disease, or Central Nervous System Infection After Craniofacial Trauma

Covariates	Number of Patients	Number of Events	Unadjusted Hazard Ratio (95% Confidence Interval)	Adjusted* Hazard Ratio (95% Confidence Interval)
Gender				
Female	185,806	2382	1.48 (1.40–1.56)	1.50 (1.42–1.59)
Male	199,705	2152	1.00 (reference)	1.00 (reference)
Age				
55–64 years	138,716	689	1.00 (reference)	1.00 (reference)
65–74 years	121,752	1,566	2.53 (2.31–2.76)	2.53 (2.31–2.77)
75–84 years	94,866	2,151	4.77 (4.38–5.20)	4.77 (4.38–5.20)
85+ years	30,177	578	4.22 (3.78–4.71)	4.06 (3.64–4.54)
History of seizure				
No	377,634	4,808	1.00 (reference)	1.00 (reference)
Yes	7877	176	1.57 (1.35–1.82)	1.37 (1.17–1.59)
History of cerebrovascular disease				
No	342,188	4,143	1.00 (reference)	1.00 (reference)
Yes	43,323	841	1.32 (1.22–1.42)	1.10 (1.02–1.18)
History of central nervous system infection				
No	383,955	4948	1.00 (reference)	1.00 (reference)
Yes	1556	36	1.58 (1.12–2.15)	1.48 (1.04–2.02)

Patients with a history of seizure, cerebrovascular disease, or central nervous system infection before head injury were excluded.

History of previously mentioned comorbidities must have occurred between the first date of traumatic brain injury or facial bone fracture and the date of dementia or December 31, 2010.

*Adjusted for sex, age, seizure history, cerebrovascular disease, and central nervous system infection.

found no significant difference between the effects of single TBI and repetitive TBI on dementia, which, nevertheless, could not prove that there were no accumulation effects of TBI on CTE or other neurodegenerative diseases, the histopathologic characteristics and neuroanatomic distribution of which are distinct from those of dementia. In addition, no operational criteria are available to confirm either a clinical or pathologic diagnosis of CTE.

Postinjury Comorbidities Versus Dementia

Our study reported that postinjury comorbidities, including PTE, cerebrovascular disease, and CNS infection, were associated with an increased risk of dementia in patients with craniofacial trauma during the follow-up period.

The association between TBI and PTE has been known since antiquity.⁴⁸ It has been estimated that TBI accounts for 10%–20% of symptomatic epilepsy in the general population and 5% of all epilepsies.^{49,50} Significant risks of epilepsy after TBIs were observed, especially the severe ones; even mild TBI seems to carry an increased long-term relative risk of epilepsy.^{51,52} Some studies have reported bidirectional effects between epilepsy and dementia.^{53–55} In the absence of other previous neurologic insult, both AD and other dementias increase the risk of generalized and partial-onset unprovoked seizures.⁵⁴ A large population-based retrospective cohort study has reported that the prevalence of AD was 5.0% in patients with epilepsy and the association was

strong (odds ratio, 2.19; 95% CI, 2.05–2.33).⁵⁵ Patients with epilepsy may bear a moderately increased risk of dementia.⁵³ However, to our knowledge, no study has shown the association between PTE and the risk of subsequent dementia in patients with craniofacial trauma. Our study indicated that patients with craniofacial trauma with PTE may be associated with an increased risk of dementia compared with those without PTE.

There is broad consensus that cerebrovascular disease, particularly the small-vessel disease that is common in aging, does not typically occur in isolation and is often associated with AD and cognitive decline.⁵⁶ Cerebrovascular disease is a major contributor to later-life dementia, accounting for up to 20% of dementia cases.⁵⁷ Approximately 10% of patients develop new-onset dementia after a first stroke, and up to one third of patients develop dementia after recurrent stroke.^{58–60} Our study indicated that the risk of dementia in patients with craniofacial trauma with subsequent cerebrovascular disease during follow-up was higher than those without comorbidities.

Cognitive impairment may present as initial symptoms at the acute stage, or as sequelae in the chronic stages, in patients with CNS infection. Many studies have shown infectious causes of dementia, including Prion disease,⁶¹ human immunodeficiency virus infection,⁶² herpes encephalitis,⁶³ neurosyphilis,⁶⁴ and bacterial meningitis.⁶⁵ Our study indicated that the risk of dementia in patients with craniofacial trauma with subsequent

CNS infection during follow-up was higher than in those without this comorbidity.

Microglia, the resident macrophages of the CNS, have been considered crucial players in the pathologic process of neurodegenerative diseases such as dementia. Microglial priming is the response of microglia to pathologic changes such as systemic inflammation or neurodegeneration in the CNS by becoming primed (i.e., becoming more susceptible to activation).⁶⁶ Neuronal damage or degeneration leads to loss or downregulation of neuronal ligands that bind to inhibitory receptors on the microglia, thereby releasing microglia from inhibition.⁶⁶ Mechanical forces from a TBI and ensuing pathophysiologic processes can compromise and disrupt neuronal circuits, leading to cerebral edema and neuroinflammation.⁶⁷ Therefore, TBI may act as a priming event for microglia.

Moreover, recent experimental and clinical evidence indicates that there are long-lasting changes in the profile of microglia after TBI. This process may set the stage for glial dysfunction and hyperactivity to subsequent challenges, such as infection, stress, ischemia, or subsequent injury.⁶⁷ Subsequent neurologic insults including trauma, infection, and ischemia may result in an exaggerated neuroinflammatory response of primed microglia.⁶⁸ This factor possibly contributes to disease progression in chronic neurodegenerative diseases. This may be the mechanism behind our study result that there is increased risk of dementia in patients who have comorbidities.

In patients with craniofacial trauma, for early identification of high-risk patients with cognitive impairment, future research is required to develop a predictive risk score for the impact of dementia. Furthermore, additional studies are warranted to identify molecular and cellular mechanisms behind microglial priming and immune reactivity and to understand the contribution of subsequent comorbidities, such as cerebrovascular disease, PTE, and CNS infection to dementia.

The limitations to the present study should be considered. First, inpatient ICD-9-CM diagnostic codes are relatively difficult for the diagnosis of dementia.⁶⁹ To overcome the misclassification bias, we restricted our inclusion to patients who were registered in the Catastrophic Illness Certificate system to avoid the false-positive cases on this side of the spectrum. Although our study may have underestimated the dementia risk, this pattern should be equal between patients with craniofacial trauma and the

general population and would not bias the magnitude of the association. Inpatient ICD-9-CM diagnostic codes were also insensitive to the diagnosis of mild TBI.⁷⁰ There would be some missed or undiagnosed mild TBI cases that were not captured in the administrative data and we might have underestimated the incidence of mild TBI. Despite this limitation, we still found higher risk of dementia in patients with mild TBI. Second, there was a possibility of reverse causality. Patients with craniofacial trauma who already had cognitive decline or undiagnosed early stage of dementia may not be captured in the administrative data.

Third, the veracity of SIR is influenced by the data limitations. Any apparent increases or decreases in dementia incidence over time may reflect changes in diagnostic methods or case reporting rather than true changes in dementia incidence: underreporting for craniofacial trauma and dementia that may not be recorded in hospitals. Moreover, SIR could not correct confounding bias except for age and sex factors.

We could not obtain the clinical information of patients with craniofacial trauma, such as the Glasgow Coma Scale, findings on computed tomography of the head, injury mechanism, and initial presentation. Moreover, there was a lack of data on family history, educational status, previous operations or illness, details of treatment during the index visit, and outpatient data.

CONCLUSIONS

A novel finding in our study was that facial bone fracture may significantly increase the risk of dementia. Other craniofacial trauma including mild TBI and moderate to severe TBI was also associated with an increased dementia risk compared with the general population in our study.

Additional studies are required to establish a causal relationship between specific facial bone fracture patterns and the future development of dementia. A clearer understanding and proper identification of high-risk groups could provide insight on appropriate prevention programs and treatment protocols.

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